Status	Finished
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Completed	Thursday, 8 October 2026, 6:44 PM
Duration	13 mins 30 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Bubble Sort
c. Linear Search
d. Binary Search

The correct answer is: Bubble Sort

Question **2**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data sequentially
b. Sorting data in a single pass
c. Rearranging data without sorting
d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **3**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Merging
b. Rearranging
c. Sorting
d. Searching

The correct answer is: Sorting

Question **4**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It uses the heap data structure
b. It bubbles up the largest element in each pass
c. It recursively sorts subproblems
d. It divides the list into sublists

The correct answer is: It bubbles up the largest element in each pass

Question **5**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470
b. 200,470,80,150,40,90,300,400,70,120
c. 80,150,200,470,40,90,300,400,70,120
d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **6**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Arranging data in a particular format
- b. Searching for data in a list
- c. Deleting data from a list
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **7**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. order()
- b. arrange()
- c. sorted()
- d. sort()

The correct answer is: sorted()

Question **8**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. order()
- b. sort()
- c. arrange()
- d. sorted()

The correct answer is: sort()

Question **9**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It requires external libraries
b. It sorts data out of the box efficiently
c. It is less efficient than custom sorting algorithms
d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **10**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is less efficient than bubble sort
b. It sorts data using a single pass
c. It does not require recursion
d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **11**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Binary Search
b. Bubble Sort
c. Merge Sort
d. Linear Search

The correct answer is: Merge Sort

Question **12**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are discarded
- c. They are split again into smaller sublists
- d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **13**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- c. A sorting algorithm is stable if it preserves the order of all keys
- d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **14**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Merging
- b. Searching
- c. Complexity
- d. Sorting

The correct answer is:
Complexity

Question **15**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower
b. It allows for faster searching
c. It increases the number of comparisons needed
d. It has no effect on the search operation

The correct answer is: It allows for faster searching